

Horn telescope discussion guide

Teacher reference

a. What is the telescope detecting?

Radio waves

b. What is the source of these radio waves? How do we know?

- The signals come from outside the earth

→ If we point the telescope at different likely sources on earth, e.g. buildings, radio

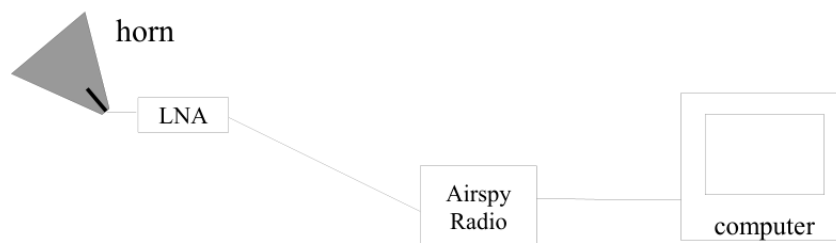
towers, other man-made structures, the “peaks” we observed do not show up. Also, the peak signals are strongest from “empty” sky, and the signal does not vary significantly over a large range of sky.

- The signals are radio waves

→ The telescope consists of an antenna (the wire in the can) and a low-noise amplifier

(LNA) designed to magnify and filter signals for the specific frequency 1420 MHz, which is in the radio frequency range.

c. Horn telescope diagram



i. Hardware

- horn & tin can
- metal wire
- cables
- amplifier
- dongle
- computer

ii. Software

- GNU Radio – open source software defined radio program